

AI OLYMPIAD

CLASS 9

The Colourful 6-Month Learning, Practice and Olympiad Preparation Guide

1 AI Basics | 2 Communicate with AI | 3 AI Applications

4 AI and Career | 5 AI Tools | 6 Responsible AI

For ages 14-15 | 6 months | 36 chapters | 6 pillars

VOLUME 1: Start-up pages + Month 1 (Chapters 1-6), expanded edition

Original educational material. Names and stories are fictional. No private data is ever required.

Hi!

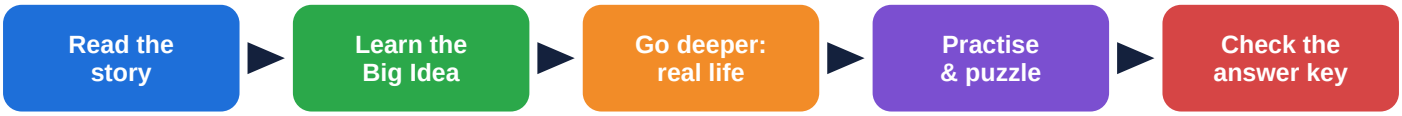
Welcome, AI Explorer

Read this first — it takes two minutes

What is this book?

This book turns the big world of Artificial Intelligence into **bite-sized, colourful, hands-on pages**. Every chapter is exactly **three pages**: one page to *learn and try*, one page to *go deeper with real-life examples*, and one page to *practise and puzzle*.

You will learn what AI is (and is **not**), write your first Python programs, see how AI helps people and the planet, explore careers, and learn to use AI **safely and honestly**.



Week	What to do	Time
Weeks 1-3	Study two chapters per week: read, do the activity, then the practice page	about 4 h
Every week	Python practice: type the code yourself, never copy-paste. Trace it by hand first!	about 1 h
Week 4	Revision, Monthly Checkpoint, mini-project and showcase	about 5 h

Golden rules of a good AI student

1. **Try first, look later.** Attempt every question before opening the answer key.
2. **Keep a mistake notebook.** Reviewing mistakes teaches more than doing new questions.
3. **AI is a helper, not the boss.** It can be wrong. Always check important facts.
4. **Be honest.** Write your own answers. Use AI to explain or check, never to do your homework for you.
5. **Stay private.** Never type your real name, address, school, phone number, photos or passwords into any AI tool.

The six-month promise

One chapter from each of the **six pillars** every month means you meet ideas, coding, real uses, careers, tools and ethics **together**. By Month 6 you will be able to explain AI clearly, write and debug Python, read data carefully, write good prompts and solve Olympiad-style questions with confidence.

Key

Your Colour Key

Every colour has a job. Learn them once, use them all year.

The six pillars each have their own colour. Look at the chapter banner to know which pillar you are in.

1. AI Basics	2. Communicate with AI	3. AI Applications
4. AI and Career	5. AI Tools	6. Responsible AI

The boxes you will meet in every chapter

Story Time

A short real-life story that shows why the topic matters.

Big Idea

The key concept in simple words, with a diagram or table.

Try It!

A safe hands-on activity. Paper, pencil and friends are enough. No internet needed.

Go Deeper and Real Life

A full page of step-by-step explanation, real-life examples, discussion questions and a mini real-life challenge.

Python Lab

Code, line-by-line explanation, expected output, common errors and a modify-it challenge.

Watch Out!

Common mistakes and misunderstandings to avoid.

Think Before You Trust

A responsible-AI habit: verify, protect privacy, be fair, be honest.

Question levels

Tag	Level	What it tests
L1	Foundation	Did you understand the idea?
L2	Application	Can you use the idea in a new situation?
L3	Olympiad Challenge	Can you reason through several steps carefully?

Safe

Safety, Setup and Offline Mode

Learn with AI without risking your privacy

Our Safety Promise

Nothing in this book asks you to share personal information. When an activity needs data, it uses **made-up** or **non-personal** data such as weather, plants or anonymous class counts.

Never share with an AI tool: real full name, home address, phone number, school name, photos of yourself or others, passwords, marks or anything a stranger could use to find you. When a program asks for a name, type a nickname.

If something online makes you uncomfortable, close it and tell a trusted adult.

Python Setup (choose one)

Computer: install a standard Python 3 from the official Python website, and use IDLE or a simple code editor. A school lab works too.

Offline alternative (no computer needed): write the code in your notebook and **trace it by hand** with a pencil, line by line. Tracing is also an Olympiad skill, so it is never wasted time.

Tip: every chapter lists the expected output so you can check your tracing.

Four things that are NOT the same

System	How it works	Learns?	Example
Ordinary program	Fixed steps written by a programmer	No	Calculator
Rule-based system	If-then rules written by people	No	Keyword spam filter
Machine learning	Finds patterns from many examples	Yes	Photo-sorting app
Generative AI	Learns patterns, then creates new text, images or code	Yes	Story-writing chatbot

Remember: automation is not the same as AI. A timer that switches on a fan is automatic, but it does not learn.

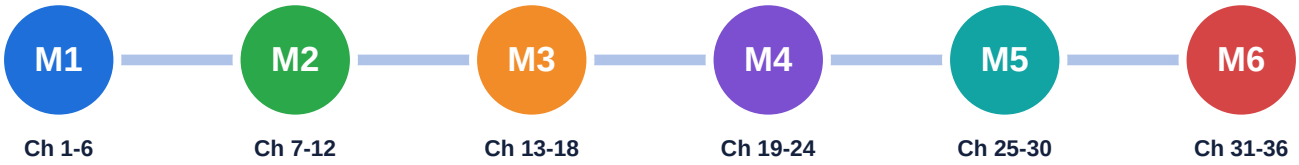
Facts, predictions and opinions

A **fact** can be checked ("water boils at 100 degrees C at sea level"). A **prediction** is a guess about the future ("robots will do every job"). An **opinion** is what someone thinks. This book keeps them separate, and so should you.

Map

Your Six-Month Map

One chapter from every pillar, every month



Month	Theme	Mini-project
1	Data, Python basics, thinking carefully	Data Diary and Climate Mini-Report
2	Patterns, decisions and loops	Health-Habit Checker (rule-based, with a safety note)
3	Generative AI, lists, fairness	Class Library Manager + fairness audit
4	Machines, prompts, verification	Prompt Workbook + verification log
5	Language models, planning, governance	Paper prototype + governance checklist
6	Machine intelligence, applications, humanity	Capstone smart app + responsible-AI report

How each month runs

Week 1: two chapters **Week 2:** two chapters **Week 3:** two chapters **Week 4:** revision, Monthly Checkpoint, mini-project.

Balance across the six months: about 35% concepts, 30% hands-on and coding, 20% responsible AI and careers, 15% exam practice.

Python skills ladder

Skill	Chapter
Variables, data types, input/output	2
Lists, first look at data summaries	1, 4, 5
Conditions, loops, basic algorithms	8
Dictionaries, functions, debugging	14
Rule-based versus learning systems	1, 7, 19
Intro to machine learning ideas	7, 31
Simple AI-inspired projects	29, 32

Menu

Contents

Page numbers show where each chapter sits in the finished 136-page book

MONTH 1 - READY in this volume

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1	From Data to Intelligence	7
2	Python in Action	10
3	AI for Our Planet	13
4	Design Your AI Future	16
5	AI Research Room	19
6	Think Before You Believe	22
	Checkpoint 1	25

MONTH 2 - coming next

Ch	Title	Pg
7	How AI Thinks with Data	26
8	Solving Problems with Code	29
9	AI for Better Health	32
10	Career Compass	35
11	AI Creator Studio	38
12	Digital Footprints and AI	41
	Checkpoint 2	44

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Ch	Title	Pg
13	Generative AI Uncovered	45
14	AI-Assisted Coding	48
15	AI at Work	51
16	Your Road to an AI Career	54
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23	AI Assistant Lab	76
24	Verify Before You Trust	79
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Ch	Title	Pg
25	Understanding Large Language Models	83
26	Prompt, Plan, Produce	86
27	AI for Enterprise	89
28	From Campus to AI Career	92
29	AI Prototype Lab	95
30	AI Governance Essentials	98
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MONTH 6 - coming next

Ch	Title	Pg
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Back of the book: Mock Olympiad papers (pp. 121-126) · Answer keys (pp. 127-132) · Glossary (pp. 133-134) · Student portfolio and certificate (pp. 135-136)

1 From Data to Intelligence

Pillar 1 · AI Basics · Learn: data, information, knowledge, and four kinds of systems

Story Time: The Canteen That Wasted Samosas

Arun's school canteen made 60 samosas daily. Some days they ran out; other days 20 were thrown away. Arun wrote 20 days of sales in a table, grouped them by weekday and by rain, and spotted a pattern: sales were higher on dry Mondays and Fridays, lower on rainy days. The manager changed her plan and wasted far less food. **No magic: raw numbers became a better decision.**

Big Idea: The Data Ladder



Example: 42, 38, 55 (data) → "Monday average is 58" (information) → "dry, busy days sell more" (knowledge) → "make more on Monday" (decision). More data is not always better: **quality, balance and context** matter.

Four systems, not one: do not call everything AI!

System	Who decides what it does?	Learns?	Example
Ordinary program	Programmer writes exact steps	No	Calculator, times-table printer
Rule-based system	Experts write if-then rules	No	Spam filter using fixed keywords
Machine learning	It finds patterns in many examples	Yes	App that sorts photos it was trained on
Generative AI	It learns from huge data, then creates new content	Yes	Chatbot that writes a story

Try It! Data Detective (10 minutes, paper only)

Record the pages you read each day for 7 days, or use this sample: **12, 8, 15, 5, 20, 10, 6**.

(a) Find the total and the average. (b) Write one sentence of *information*. (c) Write one *decision* it could help with.

Bonus: Is "pages read" numerical or categorical data? Is "favourite subject"?

Watch Out!

- **Not all software is AI.** Most programs just follow fixed instructions.
- **Two things together do not prove one causes the other.** Rain and low sales may be linked, but an exam day could be the real reason.
- **AI today is "narrow":** each system does particular tasks. A machine that can do everything a human can (AGI) does not exist today; predictions about it are opinions.

1 Go Deeper: From Data to Intelligence

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

Step 1: A number alone cannot be used. "42" could be samosas, marks, or degrees. Data becomes useful when we add **labels and context** (what, where, when).

Step 2: Organise it. Sort, group and summarise: totals, averages, highest and lowest, or a chart. This is how data becomes **information**. A calculator or a short Python program can do this, and that is not AI.

Step 3: Reason about it. Ask "why?" and "what else could explain this?" That reasoning builds **knowledge**. Then people make a **decision** and later check whether it worked.

Where does learning enter? In an ordinary program or rule-based system, a person decides every rule, for example "if sales are under 40, make fewer samosas". In **machine learning**, we show the computer many past examples (days with sales and weather) and it builds its own pattern for predicting. A person still chooses the problem, supplies the data and checks the result.

Features and labels, again: for a model that predicts "busy day or quiet day", the **features** could be weekday, rain and exams. The **label** is the known answer for each past day: busy or quiet.

Real Life Around You

Situation	Data	Information	Decision	Is it AI?
Cricket scorebook	Runs each over	Run rate	Change batting plan	No: simple arithmetic
Class test	Marks per question	Topics most missed	Revise those topics	No
Step counter	Steps per day	Weekly average	Set a new goal	No (a fixed formula)
Maps app	Speeds from many phones	Slow roads right now	Choose another route	Often uses learned patterns
Shop stock	Items sold daily	Best sellers	Reorder more	Maybe, if it learns to predict demand

Talk About It (pairs or class)

1. Why is "42 samosas" more useful than just "42"?

2. A phone suggests the next word as you type. Which of the four systems does that sound most like, and why?

3. Describe a situation where collecting more data would **not** help make a better decision.

Real-Life Challenge

Home and School Data Hunt: find **three places** where data is collected (for example attendance register, water meter, speedometer, timetable). For each, write the data, one piece of information from it, and whether a person or a learning system might use it. Do not record anything about individual people.

1 From Data to Intelligence — Practice Page

Python lab · quiz · puzzle · create

Python Lab: Turn Data into Information

```
sales = [42, 38, 55, 61, 47]
total = sum(sales)
average = total / len(sales)
print("Average:", average)
```

What it does: line 1 stores five numbers in a list; line 2 adds them; line 3 divides by how many there are (`len`); line 4 prints the result. **Output:** `Average: 48.6` **Is it AI?** No: it is an ordinary program.
Common errors: missing bracket `]` (SyntaxError); `Sum` with capital S (NameError); forgetting quotes around text.
Modify it: add a sixth day with 50 sales and find the new average. Then print the best day using `max(sales)`.

Quick Quiz — 5 Questions

1. The numbers 42, 38, 55 with no labels are best called… [L1]
A knowledge **B** data **C** a model **D** wisdom

2. A calculator adding two numbers is… [L1]
A machine learning **B** generative AI **C** AGI **D** an ordinary program

3. "If an email says WIN A PRIZE, mark it spam" is… [L2]
A a rule-based system **B** machine learning **C** generative AI **D** AGI

4. A system that improves by studying thousands of labelled examples is… [L2]
A an ordinary program **B** rule-based **C** machine learning **D** a calculator

5. Which is categorical data? [L1]
A height **B** favourite subject **C** temperature **D** pages read

Short Answers — 2 Questions

S1. Explain the difference between data and information, with one school example.
S2. Why is a washing-machine timer not AI?

Olympiad Puzzle [L3]

A kiosk records sales Mon-Sat: **50, 48, 52, 30, 29, 51**. It rained on Thursday (12 mm) and Friday (15 mm) only. (a) Find the average sales on rainy days and on dry days. (b) A student says "this proves rain causes low sales". Evaluate the claim in two sentences.

Create It!

Draw a comic strip, "A Day in the Life of a Number", from the moment it is recorded to the decision it helps make. Use the words data, information, knowledge and decision. No real names.

2 Python in Action

Pillar 2 · Communicate with AI · Learn: variables, data types, input and output

Story Time: The Scoreboard

Meera's class ran a quiz game, and keeping score on the board was messy. Her brother showed her that a computer can keep the score in a **labelled box** and update it instantly. She learned her first big idea: a program is just a list of instructions the computer follows exactly, in order, every time.

Big Idea: Variables are labelled boxes



A **variable** has a **name** and holds a **value**. You create one with **=** (read it as "gets the value"): `score = 25`. Names: letters, digits and underscore; no spaces; cannot start with a digit; **capital letters matter** (`Score` is not `score`).

Data types

Type	Meaning	Example	Python name
Whole number	Counting numbers, positive or negative	25, -3	int
Decimal	Numbers with a decimal point	12.5	float
Text	Letters and symbols inside quotes	"Ria"	str (string)
True / False	Yes-or-no answers	True	bool

Try It! Be the computer (paper only)

Draw boxes a and b. Follow these lines one by one and cross out old values:

```
a = 5    b = 3    a = a + b    print(a)
```

What is printed? What are the final values of a and b? **Then try:** `print("5" + "3")`. Why is it different?

Watch Out!

- = **stores** a value; it is not the "equals" of maths.
- Text needs quotes. `print(Hello)` fails, `print("Hello")` works.
- `input()` always gives you **text**, even when the user types a number. Convert with `int(...)` or `float(...)`.

2 Go Deeper: Python in Action

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

A program is like a recipe. The computer follows the steps exactly, from top to bottom, and never guesses what you meant. That is why spelling, capital letters and quotes matter so much.

Assignment runs right to left. In `a = a + b`, Python first works out the right side (a + b) and then stores the answer in a. So the old value of a is replaced.

Maths order applies. Multiplication and division come before addition and subtraction, and brackets come first, just like BODMAS. So `2 + 3 * 4` is 14, but `(2 + 3) * 4` is 20.

Comments help humans. A line starting with # is ignored by Python. Use comments to explain your plan. Below is a tuck-shop bill calculator:

Example: Tuck-Shop Bill

```
# Tuck-shop bill
samosa_price = 12
quantity = 3
bill = samosa_price * quantity
print("Bill:", bill)
```

Output: `Bill: 36`. **Modify it:** add tea at price 10, quantity 2, and print the total (56).

Real Life Around You

Real situation	Variable names	Type	Why
Tuck-shop bill	price, quantity, bill	int or float	Prices can have decimals
Game lives	lives, level	int	Whole numbers
Phone battery	battery_percent	int or float	Changes over time
Quiz player	player_name, passed	str, bool	Text and yes/no
ATM balance	balance	float	Money changes with each action

Talk About It (pairs or class)

1. Why can a variable name not contain a space?

2. Is the bill calculator below AI? Explain.

3. Fun fact: in Python, `"12" * 3` gives `121212`. Why do you think this happens?

Real-Life Challenge

Tuck-shop Plan: on paper, list the variables needed for a bill calculator with two items. Choose names and types, then trace the program by hand for 3 samosas at 12 and 2 teas at 10. (Answer: 56)

2 Python in Action — Practice Page

Python lab · quiz · puzzle · create

Python Lab: Greeting and Percentage

```
name = input("Nickname: ")
marks = 45
total = 50
percent = marks / total * 100
print("Hi", name)
print("Percent:", percent)

age = int(input("Age: "))
print("Next year:", age + 1)
```

What it does: `input` asks and stores text (use a **nickname**, not your real name!). Lines 2-4 calculate the percentage. `int(...)` turns text into a whole number so we can add 1.
Sample run (typing Ria, then 14): `Hi Ria` | `Percent: 90.0` | `Next year: 15`
Common errors: writing `Marks` instead of `marks` (NameError); forgetting `int()` gives TypeError when adding. **Modify it:** add a second subject and find the percentage out of 100.

Quick Quiz — 5 Questions

1. Which is a valid variable name? [L1]
A 2score **B** my score **C** my-score **D** my_score

2. What type is the value 3.5? [L1]
A int **B** str **C** float **D** bool

3. What does print(2 + 3 * 4) show? [L2]
A 20 **B** 14 **C** 24 **D** 9

4. x = 7, then x = x + 2, then print(x). Output? [L2]
A 7 **B** 2 **C** 9 **D** x+2

5. input() always returns… [L1]
A a whole number **B** text (a string) **C** a decimal **D** True or False

Short Answers — 2 Questions

S1. What is a variable?
S2. Why does print("5" + "3") show 53 and not 8?

Olympiad Puzzle [L3]

Trace by hand: `a = 4; b = a * 2; a = b - 3; b = a + b; print(a, b)`. What is printed? Show the value of a and b after every line.

Create It!

Plan a program for a school quiz scoreboard. List five variables you would need, give each a sensible name, and say which data type it holds.

3 AI for Our Planet

Pillar 3 · AI Applications · Learn: how AI helps with environmental data, and what it costs

Story Time: 5,000 Photos, Zero Time

In a wildlife reserve, cameras that switch on when something moves collect thousands of photos, and many show only waving grass. Checking each one by hand takes people days. A computer-vision model trained on labelled photos can sort most empty pictures from those with animals, so scientists spend their time on the interesting ones. **People still check the results.**

Big Idea: AI finds patterns in environmental data



AI can help to **sort wildlife photos**, **forecast weather**, **spot crop disease from leaf photos**, and **study air-quality readings**. But AI is a tool: people choose the problem, check the answers and make the decisions.

Problems, data and AI jobs

Problem	Data used	What AI does	Human role
Counting wildlife	Camera-trap photos	Sorts empty from animal photos	Checks and counts species
Weather forecast	Satellite, sensors, past weather	Finds patterns to predict	Issues warnings
Crop health	Leaf photos	Flags signs of disease	Decides treatment
Air quality	Sensor readings	Spots rising pollution	Takes action

Try It! Weather Diary (7 days)

Measure or note the daily high temperature for a week. Sample data: **31, 32, 30, 33, 34, 31, 29** (°C).

(a) Which day was hottest? (b) Find the average. (c) Why would one week not be enough to *train* a forecasting model?

Think Before You Trust

- **AI uses energy.** Running computers and data centres needs electricity. How much depends on the system, and researchers are still studying it. A good question: *is this use of AI worth its cost?*
- **AI can be wrong.** A wrong forecast or a missed animal has real effects, so human checking matters.

3 Go Deeper: AI for Our Planet

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

How does a model learn "empty photo" versus "animal photo"? People first label many photos. The model studies them and picks up patterns, such as shapes and textures that often appear when an animal is present. Then it is tested on new photos it has never seen.

Why can it go wrong? Night photos, partly hidden animals, unusual species or heavy rain may look different from its training photos. So scientists check a sample and fix mistakes.

Sensors and AI work as a team. Sensors collect numbers (temperature, rainfall, water level). Software finds trends. People decide actions such as issuing a warning.

The energy question. Training and running AI systems uses electricity. Ways to reduce this include choosing a smaller model when it is enough, not running AI when a simple method works, and using computers efficiently. Good engineers ask: *do we need AI for this at all?*

Real Life Around You

Idea	Data	Benefit	Limit
Rain and flood warnings	Rainfall, river levels, forecasts	Earlier warning	Forecasts can be wrong
Water-tank level alert	Sensor readings	Less water wasted	Needs working sensors
Plant-leaf checker	Leaf photos	Quick first guess	Confirm with an expert
Smart street lights	Light and movement sensors	Saves electricity	Rules or learning must be tested
Waste sorting by image	Photos of items	Faster sorting	Unusual items confuse it

Talk About It (pairs or class)

1. An AI forecast says a flood is likely. Who should decide what to do, and why?

2. Is a bigger AI always better for the planet? Explain.

3. What data could your school collect to reduce waste?

Real-Life Challenge

School Energy Audit: at three times during the day, count how many lights and fans are on in empty rooms (no names, no photos). Record the numbers, find the average and suggest one change. Then say how you would check that the change worked.

3 AI for Our Planet — Practice Page

Python lab · quiz · puzzle · create

Python Lab: Weather Summary (an ordinary program)

```
temps = [31, 32, 30, 33, 34, 31, 29]
print("Hottest:", max(temps))
avg = sum(temps) / len(temps)
print("Average:", round(avg, 1))
```

What it does: `max` finds the biggest value; `sum / len` gives the average (31.428...); `round(avg, 1)` keeps one decimal place. **Output:** `Hottest: 34` then `Average: 31.4`. This is an ordinary program, not AI.
Common errors: missing comma between list items; `round(avg 1)` without the comma.
Modify it: add three more days, print the coolest day with `min`.

Quick Quiz — 5 Questions

1. Which is an example of AI helping with environmental data? [L1]
A a fan with a speed switch **B** a poster about trees **C** a model sorting camera-trap photos using learned patterns **D** a battery charger

2. Which is a real environmental cost of running large AI systems? [L2]
A erasing weather records **B** electricity use of computers and data centres **C** making sensors useless **D** working only in cold weather

3. Why should humans still check AI wildlife-photo results? [L1]
A photos are not data **B** AI cannot read numbers **C** animals dislike cameras **D** AI can make mistakes

4. Which is data from an air-quality sensor? [L1]
A a poem about air **B** a cartoon cloud **C** a list of hourly pollution readings **D** a school timetable

5. A forecasting model trained on only 7 days of weather will probably be… [L2]
A unreliable, because the data is too little **B** perfect, since data is always enough **C** correct, because computers never err **D** better than one trained on years of data

Short Answers — 2 Questions

S1. Name two kinds of environmental data.
S2. Give one benefit and one cost of using AI for the environment.

Olympiad Puzzle [L3]

A reserve collects **1,000** photos. **60%** are empty. The AI labels **90%** of empty photos correctly as "empty"; the other 10% it wrongly labels "animal". It labels every animal photo as "animal". A human checks **every photo labelled "animal"**. (a) How many photos must a human check? (b) How many checks were saved compared with checking all 1,000?

Create It!

Design a poster, "Three Ways My School Can Save Energy", and add one way to **measure** whether it worked.

4 Design Your AI Future

Pillar 4 · AI and Career · Learn: the many roles in the AI world and how to explore them

Story Time: Kavya's Three Circles

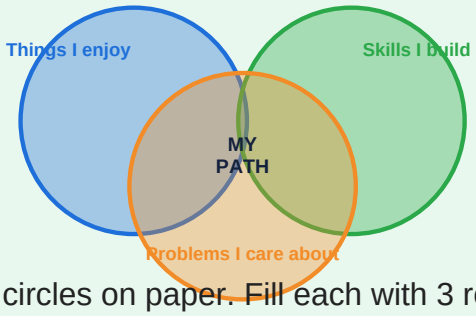
Kavya loves music and maths and wondered, "Do I have to be a programmer to work with AI?" Her teacher asked her to draw three circles: what she enjoys, what she is good at, and what problems she cares about. Where the circles met, Kavya saw careers she had never heard of.

Big Idea: AI careers are many, not one

AI needs **technical** and **non-technical** people: those who build systems, those who understand data, those who design how people use them, and experts in fields like medicine, farming or teaching who bring real-world knowledge. Skills that matter: **curiosity, maths, clear communication, teamwork, ethics, problem-solving**.

Role	What they do (in simple words)	Useful skills
Data analyst	Organises data and finds patterns that answer questions	Maths, tables, charts, care
AI / ML engineer	Builds and tests systems that learn from data	Programming, maths
Researcher	Asks new questions and tests new ideas carefully	Curiosity, evidence, writing
Product / UX designer	Makes AI tools easy and safe for people to use	Empathy, design, testing
AI ethics / policy worker	Checks fairness, privacy and safety; helps write rules	Reasoning, communication
Domain expert using AI	A doctor, farmer or teacher who uses AI tools well	Their field plus AI literacy

Try It! Draw your Interest Map



Copy the three circles on paper. Fill each with 3 real examples about you. In the middle, write careers or study ideas that fit. Finish: *"In the next 6 months I will…"* (one small, realistic goal).

Watch Out! Facts, predictions, opinions

Nobody knows exactly which jobs will exist in 20 years. Statements like "AI will take every job" or "AI will take no jobs" are **predictions**, not facts. Learn skills that stay useful: thinking, communicating and learning how to learn.

4 Go Deeper: Design Your AI Future

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

Careers are built in steps. School marks and good habits come first, then skills, projects and teamwork, and later specialised study or training. You do not need to decide your whole life now. You need to **explore**.

Be "T-shaped". A T has a wide top bar (broad skills: reading, maths, communication, teamwork) and one deep stem (one area you really enjoy). Employers and teams value both.

Real projects use teams. Imagine a team building a plant-disease app for farmers. Each person adds something different. Nobody can do it all alone.

You already use these skills. Maths class builds data skills; group projects build teamwork; writing builds communication; asking "why?" builds research.

Real Life Around You

Team member	Their part in the plant-disease app	Skill used
Agriculture expert	Explains which diseases matter and how to confirm them	Field knowledge
Data analyst	Organises leaf photos and checks data quality	Maths, care
AI / ML engineer	Builds and tests the model	Programming
Designer	Makes the app easy for farmers to use	Empathy, design
Ethics / safety reviewer	Checks privacy, fairness and clear warnings	Reasoning, communication

Talk About It (pairs or class)

1. Which role in the table would you most like to try, and why?

2. Which one skill will you build in the next month?

3. Why is "learning how to learn" important if jobs keep changing?

Real-Life Challenge

Work Story: with a parent or teacher's permission, ask one adult how technology has changed their work. Write 5 lines: what changed, one new skill they learned, and one thing that stayed the same.

4 Design Your AI Future — Practice Page

Python lab · quiz · puzzle · create

Python Lab: A First List

```
interests = ["music", "maths", "art"]
print(len(interests))
print(interests[0])
interests.append("coding")
print(interests)
```

What it does: a **list** keeps items in order inside square brackets. `len` counts them; `interests[0]` is the **first** item (counting starts at 0); `append` adds one to the end. **Output:** 3, `music`, `['music', 'maths', 'art', 'coding']`. **Common errors:** `interests[3]` on a 3-item list gives IndexError (valid numbers are 0, 1, 2). **Modify it:** add two of your own interests and print the last item with `interests[-1]`.

Quick Quiz — 5 Questions

1. Which role mostly uses data to find patterns and answer questions? [L1]
A network cable installer **B** data analyst **C** furniture designer **D** bus driver

2. A doctor learns to use an AI tool to help read scans. This shows that… [L1]
A only programmers need AI skills **B** AI will replace all doctors tomorrow **C** AI skills are useful in many fields **D** doctors must stop studying medicine

3. Which statement is a prediction, not an established fact? [L2]
A Python is a programming language **B** Data analysts work with data **C** Software is used in many hospitals **D** Every job will be done by AI within 20 years

4. A good way to start exploring AI at Class 9 is to… [L1]
A try small safe projects and learn basics step by step **B** wait until college to learn anything **C** avoid maths and reading **D** share passwords to join online clubs

5. If interests = ["music","maths","art"], then interests[0] is… [L2]
A "maths" **B** "art" **C** "music" **D** 3

Short Answers — 2 Questions

S1. Name three skills useful for AI careers that are not coding.
S2. What is the difference between a fact and a prediction about jobs?

Olympiad Puzzle [L3]

Asha, Bala and Chitra each want a different role: **data analyst, researcher, designer**. Clues: (1) the designer enjoys drawing; (2) Asha and Bala both dislike drawing; (3) Bala is not the researcher. Who wants which role? Explain each step.

Create It!

Write a 5-line letter from "Future Me" (one year from now) describing something you learned about AI. Use no private details.

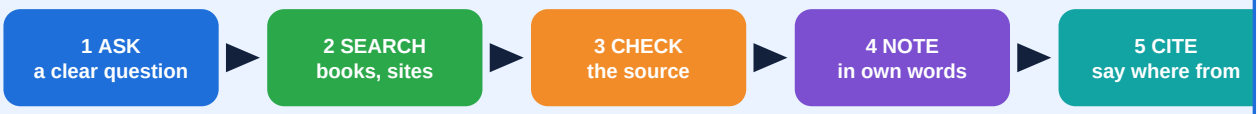
5 AI Research Room

Pillar 5 · AI Tools · Learn: researching properly, using AI as a helper, and checking sources

Story Time: "Where Did You Get This?"

Rohan had to write about solar energy. A chatbot gave him a neat paragraph, and he pasted it in. His teacher asked, "Where did you get this?" and he could not say. Worse, one fact was wrong. Rohan learned to research step by step, and to use AI as a **helper for ideas**, not as the final source.

Big Idea: The Research Route



Use the **WHO-WHEN-WHY-PROOF** check: **Who** wrote it? **When** was it written? **Why** was it written (teach, sell, persuade)? Is there **proof** and do other reliable sources agree?

Source	Good for	Watch out
School textbook / encyclopaedia	Reliable basic facts	May be a few years old
University or government website	Data and expert explanations	Check the address and date
News article	Recent events	Check who wrote it and other reports
Social media post	Finding leads and opinions	Often wrong or missing context
AI chatbot answer	Ideas, simple explanations, practice questions	Can be wrong or invent facts and sources

Try It! Source Detective (offline-friendly)

Pick one question: *"How does a solar panel make electricity?"* Find **two sources** (a book plus a second book, magazine or site). Write the author, year and title for each. Write **three sentences in your own words**. Do your two sources agree? If not, what would you check next?

Think Before You Trust

- **AI may invent references.** A book title or web address that sounds real might not exist. Always check that it does.
- **Academic honesty:** copying text word for word without credit is **plagiarism**. Put ideas in your own words and name your sources.
- **Privacy:** never paste personal details or private documents into an AI tool.

5 Go Deeper: AI Research Room

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

Search well. Use specific keywords ("solar panel how it works") rather than a whole sentence full of extra words. Open more than one result.

What AI is good for in research: explaining a hard idea in simple words, suggesting keywords, brainstorming questions, and quizzing you after you studied. **What to avoid:** treating its answer as a final fact, trusting references without checking, or submitting its text as your own.

Take 3-column notes. Column 1: the fact. Column 2: where it came from (author, year, title). Column 3: your own words. This makes citing easy and avoids accidental copying.

A simple citation can be: Author, Year, Title, Where you found it. Teachers may ask for a specific style, so follow theirs.

Good use of AI	Risky use of AI
"Explain photosynthesis simply, then ask me 3 practice questions."	"Write my full science essay."
"Suggest 5 keywords to search about solar energy."	Copying references without checking they exist.
"Here is my own paragraph; where is it unclear?" (no private details)	Pasting personal or private information.

Real Life Around You

Who checks sources?	How	Lesson for you
Journalists	Confirm with several sources before publishing	Use more than one source
Doctors	Follow current guidelines, not a single article	Prefer up-to-date, expert sources
Engineers	Read official manuals and test designs	Test and verify, do not assume
Shoppers	Compare many reviews, not just one	One opinion can mislead

Talk About It (pairs or class)

1. Why might two sources give different numbers for the same fact? What would you check?

2. Write one **good** way and one **risky** way to use an AI tool for a school project.

3. Why is writing notes in your own words better than copying and pasting?

Real-Life Challenge

Build a 3-column notes table (Fact | Source | My words) for the question "How does a solar panel make electricity?" with at least three rows from two different sources.

5 AI Research Room — Practice Page

Python lab · quiz · puzzle · create

Python Lab: Research Log

```
sources = ["Textbook", "Encyclopedia"]
print("Sources found:", len(sources))
sources.append("Science magazine")
print("Now:", len(sources))
print(sources)
```

What it does: keeps a list of the sources you found and counts them. `append` adds a new one. **Output:** `Sources found: 2`, then `Now: 3`, then `['Textbook', 'Encyclopedia', 'Science magazine']`.
Common errors: forgetting quotes around the names; writing `sources.add` (lists use `append`). **Modify it:** add a "Date" idea: build a second list with the years of each source.

Quick Quiz — 5 Questions

1. What is the best first step of research? [L1]
A copy the first search result **B** write a clear question **C** print many pages **D** ask AI to write the final answer

2. Which is usually most reliable for school science facts? [L1]
A an anonymous undated post **B** a viral video with no source **C** a current school science textbook **D** a friend's guess

3. A chatbot gives a book title and author. What should you do? [L1]
A check that the book really exists in a library catalogue or publisher site **B** put it in your list now, it sounds exact **C** ask the chatbot to promise it is real **D** ignore all books forever

4. Copying a paragraph word for word without credit is… [L1]
A brainstorming **B** paraphrasing **C** summarising **D** plagiarism

5. Writing notes in your own words helps because… [L1]
A they are always longer **B** it shows you understood and avoids copying **C** it removes the need for sources **D** it makes facts true

Short Answers — 2 Questions

S1. List four steps of the research route.
S2. Why should you not rely on a single source?

Olympiad Puzzle [L3]

A student checks **5** claims from an AI answer. **3** are confirmed by two good sources, **1** cannot be found anywhere, **1** is contradicted by two good sources. (a) What percentage was verified? (b) If an AI answer has 20 claims and the same rate holds, how many would need to be rechecked or removed?

Create It!

Design a "Source Detective" bookmark with the WHO-WHEN-WHY-PROOF check and one picture or symbol for each letter.

6 Think Before You Believe

Pillar 6 · Responsible AI · Learn: hallucinations, misinformation and the STOP habit

Story Time: The Message That Closed "All Schools"

One evening a message spread: "Government has ordered ALL schools closed tomorrow! Forward to everyone!" Hundreds of students shared it within an hour. It was false, and many arrived the next morning confused. Nobody had checked who sent the original.

Big Idea: AI hallucinations and misinformation

An **AI hallucination** is when an AI gives an answer that is **false or made-up but sounds convincing**. It happens because language AI predicts likely words; it does not automatically check facts. **Misinformation** is false information shared by mistake; **disinformation** is false information created or spread on purpose.



Warning sign	Example	What to do
Shocking or scary claim	"Everything closes tomorrow!"	Pause and look for an official source
No author or date	Anonymous forward	Treat as unverified
Perfect-sounding answer, no source	Chatbot gives exact numbers but no reference	Search two reliable sources
"Share with everyone!"	Chain messages	Do not share until checked

Try It! Fact or Fib? (pairs, 15 minutes)

Decide **True or False**, then check each in a textbook or encyclopaedia and note your source: (1) Water boils at 100 °C at normal sea-level pressure. (2) The Great Wall of China can be seen with the naked eye from the Moon. (3) Humans use only 10% of their brains. (4) Sound travels faster in water than in air. (5) Lightning never strikes the same place twice.

Think Before You Trust

- A **confident tone is not evidence.** Chatbots can sound sure and still be wrong.
- Use AI to **get started**; use reliable sources to **confirm**.
- **Never type** your address, phone number, school name, photos or passwords into an AI tool.

6 Go Deeper: Think Before You Believe

Explanation · real life around you · talk it over · challenge

Explain It Step by Step

Why do false messages spread? They are often shocking or emotional, they arrive from friends we trust, and sharing takes one tap. Checking takes a little longer, so many people skip it.

Kinds of misleading content: an honest mistake; an old photo shown with a new story; a real quote cut short; a joke taken seriously; an edited image; or something invented on purpose.

Deepfakes are images, audio or video made or changed by AI so that they look real. Warning signs can include strange details, no original source, or a claim that seems designed to shock. But looks alone are not proof, so **find the original source**.

Scam links such as "free recharge, click now" try to rush you. Do not click, do not enter personal details, and tell a trusted adult.

Real Life Around You

Situation	Why risky	Better action
Forwarded health tip with no source	Wrong health advice can harm	Ask a doctor or check an official health site
"Free gift, click now!"	Possible scam to steal details	Do not click; tell an adult
Old flood photo with a new claim	Real photo, wrong story	Search for the original photo source
Screenshot of a famous "quote"	Screenshots are easy to fake	Find the original speech or article
AI answer with exact numbers, no source	May be invented	Check two reliable sources

Talk About It (pairs or class)

1. Have you ever almost shared something false? What made it believable?
2. Why is "it looks real" not enough to prove a video is real?
3. What is one kind thing you can do when a friend shares a false message?

Real-Life Challenge

STOP Test: compare one claim from your textbook with one rumour you have heard (no names). Apply Stop, Think, Other sources, Prove. Write what you found and how sure you are.

6 Think Before You Believe — Practice Page

Python lab · quiz · puzzle · create

Python Lab: Claim Checker

```
verified = 3
total = 5
percent = verified / total * 100
print("Verified:", percent, "%")
```

What it does: works out what percentage of claims you have verified. **Output:** `Verified: 60.0 %`. Python puts a space between items separated by commas inside `print`.
Common errors: forgetting the closing quote on "%"; dividing by 0 if `total` is 0 (ZeroDivisionError). **Modify it:** change `verified` to 4 and print the new percentage.

Quick Quiz — 5 Questions

1. An AI hallucination is… [L1]
A a computer virus **B** a false or made-up answer that sounds convincing **C** a slow internet link **D** a blurry photo

2. Why can a chatbot hallucinate? [L1]
A it predicts likely text and does not automatically check facts **B** it only copies one book **C** it has feelings **D** it never uses data

3. Before forwarding a shocking message you should… [L2]
A forward it quickly to warn everyone **B** delete your phone **C** check reliable sources first **D** add more dramatic words

4. Disinformation is… [L1]
A an honest mistake **B** false information created or spread on purpose **C** true information **D** a school notice

5. Which should NOT be typed into a chatbot? [L2]
A a maths question **B** a general science question **C** your home address and phone number **D** a request to explain photosynthesis

Short Answers — 2 Questions

S1. Explain the STOP habit in your own words.
S2. Why does a confident tone not prove that an answer is true?

Olympiad Puzzle [L3]

A rumour spreads: on Day 0 one person knows. Every day, each person who knows tells **3 new people** (nobody is told twice). How many people know the rumour by the **end of Day 3**, counting the first person? What does this tell you about checking before sharing?

Create It!

Make a poster, "Pause Before You Share", with the STOP steps and one example from your own (non-private) experience.

1 Monthly Checkpoint — Month 1

Eight questions, one project, one checklist

Checkpoint Quiz

1. Which is information, not raw data? [L1]
A 31 **B** Mon **C** Average temperature this week was 31.4°C **D** rain

2. x = 3, y = x * 2, x = y + 1, print(x, y). Output? [L2]
A 7 6 **B** 4 6 **C** 7 7 **D** 3 6

3. A photo app learns to group pictures after seeing many examples. It is… [L1]
A an ordinary program **B** rule-based **C** a fixed timetable **D** machine learning

4. Which can make a model learn wrong patterns? [L2]
A unrepresentative or wrong data **B** clean data **C** correct labels **D** large balanced data

5. Best first check of a surprising claim: [L2]
A count the likes **B** look for it in two reliable sources **C** trust it if it sounds confident **D** ask a friend to guess

6. Which is an opinion/prediction? [L2]
A water boils at 100°C at sea level **B** AI will replace all teachers within 10 years **C** a list is a Python data type **D** a sensor produces data

7. What type is "25" (with quotes)? [L1]
A int **B** float **C** str **D** bool

8. temps = [4, 9, 6]; print(max(temps)). Output? [L1]
A 4 **B** 6 **C** 19 **D** 9

Mini-Project: Data Diary and Climate Mini-Report

1. Choose a safe, non-personal measure (daily temperature, rainfall, cups of water used at home). **2.** Record it for 10 days in a table. **3.** Find total, average, highest and lowest (by hand or with the Python lab code). **4.** Draw a bar chart by hand. **5.** Write: one sentence of *information*, one cautious *conclusion* and one *limit* of your data (for example "only 10 days"). **6.** Present on one page.
Self-score (20): data collected honestly /4 · calculations correct /4 · clear chart /4 · cautious conclusion /4 · limits explained /4.

Progress Checklist (tick when you are sure)

[ ] I can define data, information and knowledge.
[ ] I can tell ordinary programs, rule-based systems, machine learning and generative AI apart.
[ ] I can write and trace a Python program with variables, input and output.
[ ] I can make and use a simple list.
[ ] I can explain two ways AI helps the environment and one cost.
[ ] I can name three AI-related careers and the skills they need.
[ ] I can check a source with WHO-WHEN-WHY-PROOF.
[ ] I can explain an AI hallucination and use the STOP habit.

Key

Answer Key — Month 1

Try first! Then check. (In the finished book, keys sit together on pages 127-132.)

Ch 1	MCQ: B D A C B. Try It: total 76, average about 10.9. Python: 48.6. S1: data are raw facts (42); information is organised data (average 48.6). S2: it follows fixed steps and learns nothing. Puzzle: rainy (30+29)/2 = 29.5; dry (50+48+52+51)/4 = 50.25. The claim is too strong: only 2 rainy days, other causes (exam, holiday) possible, so it shows association, not proof.
Ch 2	MCQ: D C B C B. Try It: a=8 printed; b stays 3; "5"+"3" gives 53 because both are text and + joins them. Python: Hi Ria / Percent: 90.0 / Next year: 15. S1: a named place that stores a value. S2: they are strings, so + joins text. Puzzle: a=4; b=8; a=5; b=13. Prints **5 13**.
Ch 3	MCQ: C B D C A. Try It: hottest 34 (day 5); average 31.4. Python: Hottest: 34 / Average: 31.4. S1: any two of sensor readings, camera or satellite images, weather records. S2: e.g. saves sorting time; uses electricity. Puzzle: empty photos = 600, wrongly labelled "animal" = 60; animal photos = 400. (a) **460** checks. (b) **540** saved.
Ch 4	MCQ: B C D A C. Python: 3, music, ['music', 'maths', 'art', 'coding']. S1: e.g. communication, teamwork, ethics, maths, curiosity. S2: a fact can be checked; a prediction is a guess about the future. Puzzle: designer enjoys drawing, Asha and Bala dislike it, so **Chitra = designer**. Bala is not the researcher, so **Bala = data analyst**, **Asha = researcher**.
Ch 5	MCQ: B C A D B. Python: 2, 3, then the list of three sources. S1: ask, search, check, note in own words, cite (any four). S2: one source may be wrong or biased. Puzzle: (a) 3 of 5 = **60%**. (b) 40% of 20 = **8** claims to recheck or remove.
Ch 6	MCQ: B A C B C. Fact or Fib: 1 True, 2 False, 3 False, 4 True, 5 False (tall structures are often struck many times). Python: Verified: 60.0 %. S1: Stop, Think who made it, check Other sources, Prove before sharing. S2: a tone is style, not evidence. Puzzle: 1 + 3 + 9 + 27 = **40**; spread grows very fast, so check before sharing.
Test 1	MCQ: C A D A B B C D. (Q2: y=6, then x=7, so "7 6". Q8: the largest of 4, 9, 6 is 9.) Score: 7-8 great; 5-6 revise weak chapters; below 5 re-read and retry the Try It! activities.

Creative tasks: what a strong answer shows

Ch 1: all four ladder words used correctly. Ch 2: sensible variable names and correct types. Ch 3: an energy-saving idea plus a way to measure it. Ch 4: a realistic goal, no private details. Ch 5: each letter of WHO-WHEN-WHY-PROOF explained. Ch 6: STOP steps and a real-life example without names.